

TOP DOWN DESIGN

Agenda

- Programming and implementation issues
- System design techniques
- Programming techniques
- Modular designs of a program
- Top down design & Stepwise refinement
- References

Programming and Implementation issues

What is Program?

- A computer program is a collection of instructions that performs a specific task when executed by a computer.
- A computer requires programs to function, and typically executes the program's instructions in a CPU.

What is programming language?

- A programming language is a formal computer language designed to communicate instruction to a machine, particularly a computer.
- Programming language can be used to create programs to control the behavior of a machine or to express algorithms.

System Design Techniques

- Sequence of computational techniques to solve a problem is called system design.
- 2 types of system design are:
 - ✓ Top down design
 - ✓ Bottom up design

System Design Techniques

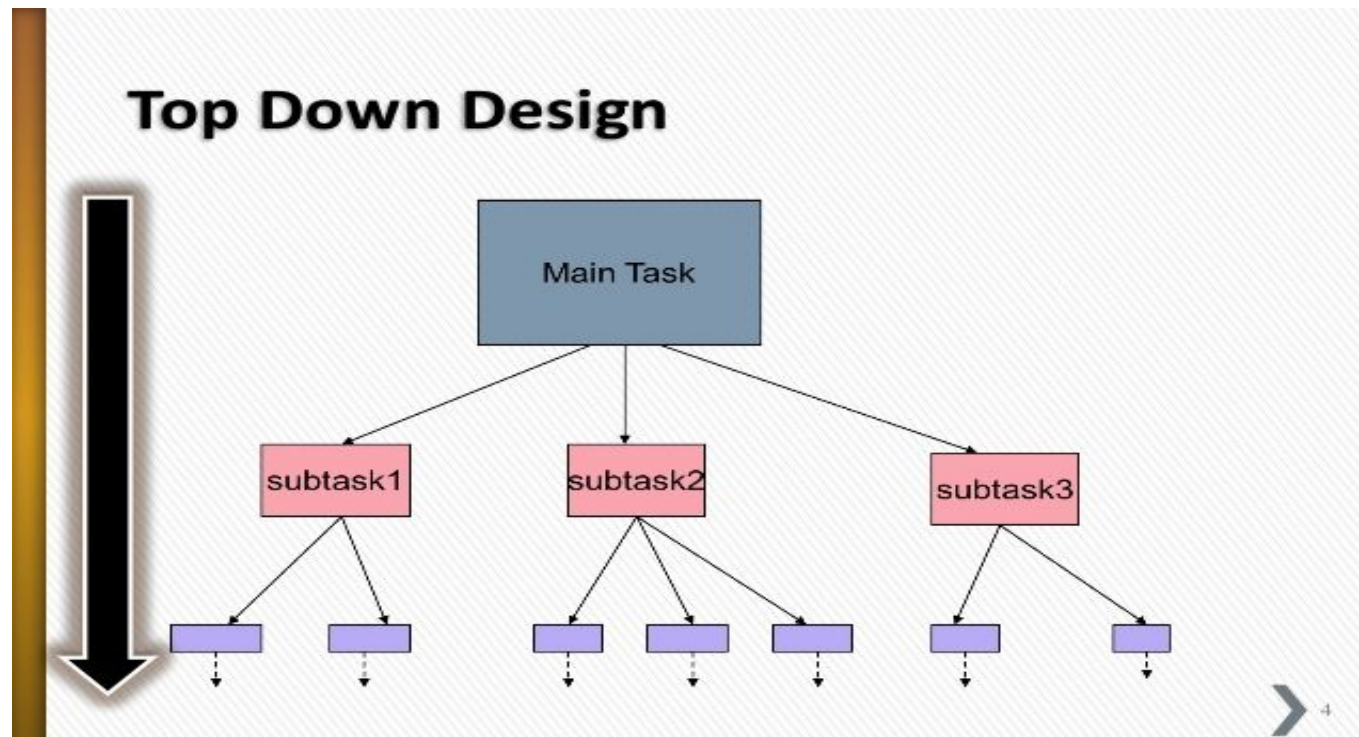
Top down design

- Best suited for solving big problem.
- Big problem is divided into a number of small problems which can be managed easily.

Bottom up design

- Opposite of top down approach
- Basic subroutines are written first and those subroutines are used to make sophisticated subroutines.

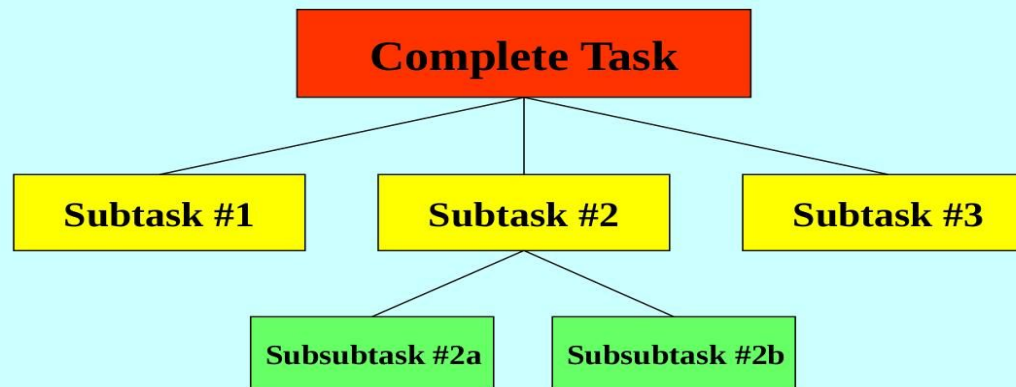
Top down design & Stepwise refinement



Top down design & Stepwise refinement

Stepwise Refinement

- The most effective way to solve a complex problem is to break it down into successively simpler subproblems.
- You start by breaking the whole task down into simpler parts.
- Some of those tasks may themselves need subdivision.
- This process is called *stepwise refinement* or *decomposition*.



Programming Techniques

- 2 techniques for programming
 - ✓ Linear programming
 - ✓ Structured programming

Programming Techniques

□ Linear programming

- Similar to sequential programming.
- Doesn't have any decision making.

□ Structured programming

- Includes conditional flow.
- Branching and looping.
- Helps to reduce errors in writing complex programs.
- Whole program is broken into modules and sub-modules.
- Programs can be written more easily and quickly.
- Easy to debug.
- High productivity.

Basic Constructs of Structured Programming

The three constructs in structured programming are:

- Sequence
- Selection
- Iteration

Basic Constructs of Structured Programming

Sequence

- It is a continuous set of actions where one is followed by another till the expected result is obtained.

Statement 1

Statement 2

Statement 3

...

Statement n

Basic Constructs of Structured Programming

Selection

- It is used in conditional execution.
- When you take some action based on the results of logical test, that is a selection construct.

If (condition)

 True statements;

Else

 False statements;

Endif

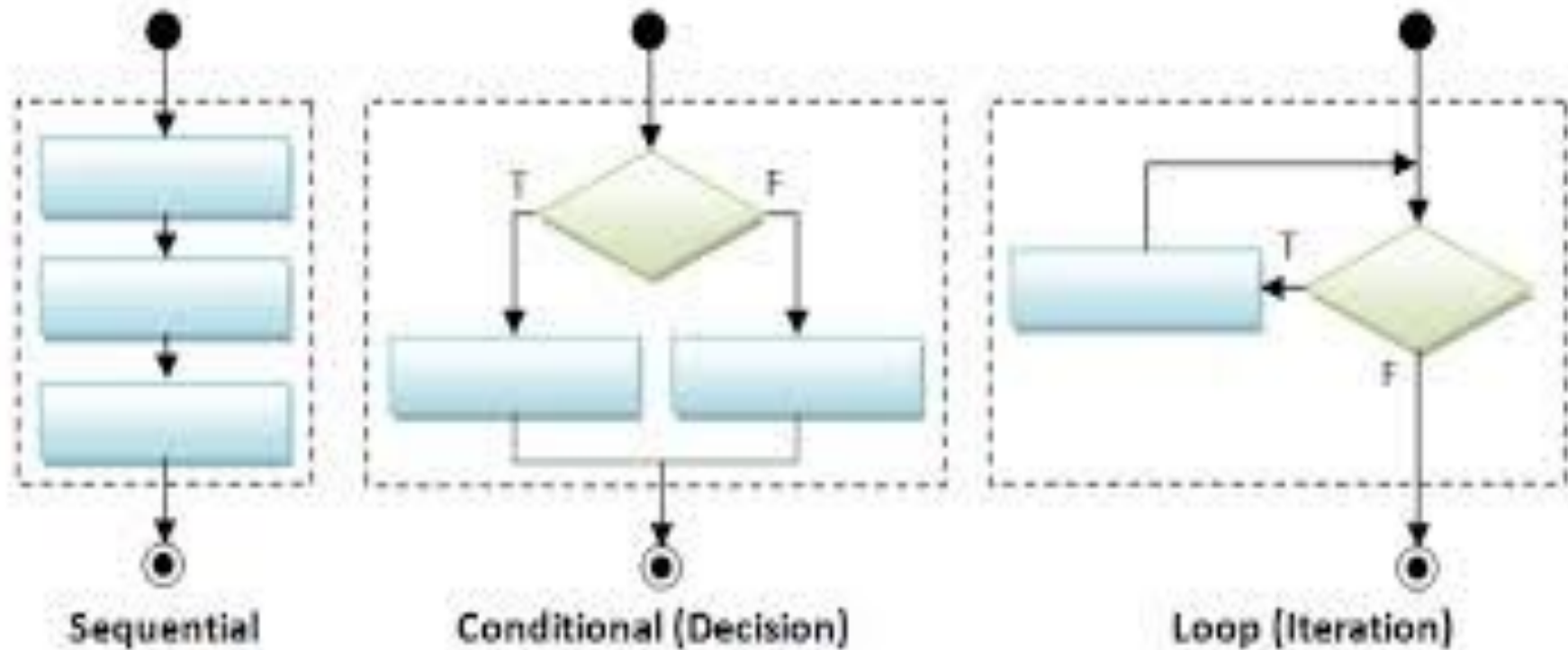
Basic Constructs of Structured Programming

Iteration

- Iteration is repetitive execution of consecutive instructions.
- It is also called looping.
- It may either be conditional or unconditional looping.



Basic Constructs of Structured Programming



Modular designs of a program

- Each program consists of various modules and each module performs separate functions.
- Requirements for modular design are:
 - ✓ Coupling
 - ✓ Cohesion
 - ✓ Size
 - ✓ Shared use

THANK YOU